MATH2056 Spring-2020, F Date: 3/7/20

The books, GSM phones and calculators are not allowed. The lecture notes can be used. All the answers must be clearly stated, otherwise no partial credit will be given.The duration is 2 days..

Important: you will have ZERO points for your answers containing the same errors with others, regardless of the nature of the error.

You have to answer the following questions:

1. A pandemic infection is spreading in a population and it is killing 10% of sick people in a month. The rest of sick people are recovering as of the end of the month. According to statistics, each month, the 5% of people, who are well are getting sick. Unfortunately, there is no any evidence to say that the people, who are recovered will develop an immune response. Answer the following questions according to information given.

    a. Determine the state transition matrix A. Explain your reason. (10 pts)
    b. Find the characteristic equation. (10 pts)
    c. Calculate the eigen values and eigen vectors for the process. (15 pts)
    d. Propose an initial condition the reach the stady state in shortest duration. (20 pts)
    e. Assume all the population are well initially, determine the number of death people at 10th month? (20 pts)

If you cannot determine the state transition matrix A, please use the following matrix to continue to solve the question:

$$\begin{bmatrix} \mathbf{1} & \mathbf{-0.1} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \mathbf{-1} \\ \mathbf{0} & \mathbf{0.1} & \mathbf{0} \end{bmatrix}$$

2. Suppose S is a fixed invertible 3 by 3 matrix. This question is about all the matrices that are diagonalized by S, so that $S^{-1}AS$ is diagonal. Comment about the space formed by 'A matrices', is it a vector space or not? (15 pts)

3. Let $p(t) = (t-1)^2$ and $q(t) = (t+1)^2$. Find the norms $\|p\|$ and $\|q\|$, the angle between p and q, and the distance between p and q **in the interval of -1<t<0.** (20 pts)

4. Evaluate

$$\begin{vmatrix} 1 & 2 & 2 & 0 & 3 \\ 0 & 1 & 2 & 0 & -1 \\ 2 & 3 & 5 & 2 & 1 \\ 1 & 1 & 2 & 1 & 1 \\ 1 & 3 & 3 & 0 & 4 \end{vmatrix}.$$ (10 pts)

You can choose 2 questions to answer from the following set:

5. In R^3 with the standard inner product find the distance between the vector u and the plane spanned by v1 and v2, where $\mathbf{u} = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}$, $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$. (5 pts)

6. Consider the linear equation set given below. Here α,β and θ are scalar numbers. Find α,β and θ values **for each case** in order to satisfy the the conditions given. (15 pts)

   a) The equation set should have infinite solutions and the solution space is one dimensional.
   b) The equation set has a soultion for any value of θ.

$$\frac{x}{\alpha} + y + \alpha^2 z = 2$$

$$\beta x + y + \alpha z = 5\theta$$

$$x + \beta y + \alpha z = \beta$$

7. Find the rank and bases for the row, column, and null spaces for the following matrix. (15 pts)

$$A = \begin{bmatrix} -2 & 2 & 3 & 7 & 1 \\ -2 & 2 & 4 & 8 & 0 \\ -3 & 3 & 2 & 8 & 4 \\ 4 & -2 & 1 & 5 & -7 \end{bmatrix}$$